



Credit: NATO

A research vessel (1), along with surface buoys (2), a submarine (3), UUVs (4) and embedded underwater sensors (5).

swarm of UUVs, and capable of spending months in the ocean on its own. It has a nautical range of 6,500 nautical miles on a single charge, and does not require a host ship for operation. It has a modular design that can accommodate a variety of payloads and can even house payloads (weapons) on its exterior. Both of these facets point to its potential for being armed. Russia has a nuclear powered and nuclear armed UUV capable of long term underwater surveillance. The Poseidon was first revealed through leaked Pentagon material in 2015, with Putin formally announcing its existence along with six other futuristic superweapons in 2018. The drone also uses advanced stealth technology to avoid detection by tracking devices, by moving forward slowly and silently through the water. Poseidon has the capability of attacking US port cities, and can contaminate wide swathes of water with radiation.

India too is working on a number of AUVs, with the eventual aim of replacing submarines with drones. One of the many models being developed by the DRDO is capable of being launched from a submarine torpedo, a state of the art submadrone. After completing its mission, it is designed

to drop onto the seafloor for subsequent recovery by another underwater drone. DRDO is also developing AUVs with private companies in India and international corporations as well, for a number of ISR applications. Improved underwater covert surveillance capabilities can potentially boost the adventurism of the Navy, increasing their range of operations.

The Centre for Maritime

Research & Experimentation (CMRE) has provided NATO forces with an entire underwater ecosystem to gather underwater intelligence and coordinate with land-based sources. A warship uses conventional wireless signals to communicate with surface buoys that act as gateways to in turn relay signals to UUVs and submarines using acoustic modems. The Submarines also have the capability of communicating with the UUVs. Embedded on the seafloor are sensors that gather data over long periods of time, which are then relayed up the chain to either the UUVs or the surface buoys. The “research vessel” can then communicate back with the land or the submarine for coordinating further actions.

There are many reasons to choose autonomous armed platforms over conventional ones. The primary one is to outdo the competition, which is resulting in an arms race. The Australian Army Research centre has a roadmap towards developing fully autonomous robotic and armed systems (RAS), which will

be deployed after semi-autonomous systems that can be optimally crewed. The U.S. Army Robotic and Autonomous Systems Strategy published in 2017 states, “The Army must pursue RAS capabilities with urgency because adversaries are developing and employing a broad range of advanced RAS technologies as well as employing new tactics to disrupt U.S. military strengths and exploit perceived weaknesses. RAS are increasingly important to ensuring freedom of maneuver and mission accomplishment with the least possible risk to Soldiers.” Russia under Putin is aggressively pursuing the development of “intelligent” weaponized drones. This autonomy race is a cause for global concern as history shows that in an effort to outdo the competition, countries may build up large and destabilizing stockpiles of weapons, leading to long running standoffs, and threatening global peace, similar to the Cold War. When it comes to AUVs, there is a clear cost benefit as entire fleets of drones can be deployed at the same cost of one modern destroyer, and not only would these robotic swarms be harder to target, more of them can be lost to enemy action while the fleet still retains most of its capability. There is also the question of the



Credit: Boeing

The Boeing Echo Voyager

cost of training crew, and keeping them fed and housed in a hostile environment, and the limits on for how long such crews can remain on underwater missions. For say ocean surveillance of hostile submarines, drones are just the cheaper and better option than manned submarines patrolling the waters. **d**